

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285524

Kind Code

A1

Publication Date

September 11, 2025

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ACTION MONITORING SYSTEM AND ACTION MONITORING METHOD

Abstract

An action monitoring system is configured to monitor an action of a monitoring target person living in a residence, and includes occupancy sensors, a storage section, and a determination section. The occupancy sensors are provided in places different from each other in the residence. The determination section is configured to store, in the storage section in advance, a specific action defined by at least a reaction order of the multiple occupancy sensors, and store a reaction time as reaction data in the storage section when any of the multiple occupancy sensors reacts. The determination section is further configured to extract a current action based on the reaction data stored in the storage section, and compare the current action with the specific action to determine whether the specific action has been normally performed.

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Appl. No.: 18/697613

Filed (or PCT Filed): October 06, 2021

PCT No.: PCT/JP2021/036986

Publication Classification

Int. Cl.: G08B21/04 (20060101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD

[0001] The present description discloses an action monitoring system and an action monitoring method.

BACKGROUND ART

[0002] Conventionally, as this type of action monitoring system, there has been proposed an action monitoring system that collects a daily life pattern of a watching target person using a sensor such as an occupancy sensor installed in a residence and compares the daily life pattern with a current action pattern to obtain a state of the current action pattern (for example, see Patent Literature 1). In this system, as the daily life pattern, a start time, duration, the number of times, and the like of each event such as wake-up, toilet, meal, bathing, and sleep are read from the sensor, a degree of suitability of the lifestyle action based on the read start time, duration, and number of times of occurrence is obtained, and when the degree of suitability is smaller than a predetermined value, it is determined that the life pattern is abnormal. For example, a case where a user is sleeping in a normal meal time, a case where the number of times visiting toilet is small, or a case where a user is sleeping for a long time can be exemplified.

Patent Literature

[0003] Patent Literature 1: JP-A-2014-106636

SUMMARY OF THE INVENTION

Technical Problem

[0004] However, the system described above is limited to a system capable of determining whether an action is normal based on the presence or absence of the target person in one area such as wake-up, toilet, meal, bathing, and sleep, and there is no reference to determining whether an action moving across multiple areas (rooms), such as cleaning or garbage disposal, is normal.

[0005] A main object of the present disclosure is to provide an action monitoring system and an action monitoring method capable of appropriately monitoring an action of moving across multiple areas.

Solution to Problem

[0006] The present disclosure employs the following means in order to achieve the main object described above.

[0007] A gist of an action monitoring system of the present disclosure is an action monitoring system configured to monitor an action of a monitoring target person living in a residence, the action monitoring system including multiple occupancy sensors provided in places different from each other in the residence and configured to detect the monitoring target person, a storage section configured to store data, and a determination section configured to store, in the storage section in advance, a specific action defined by at least a reaction order of the multiple occupancy sensors, store a reaction time as reaction data in the storage section when any of the multiple occupancy sensors reacts, extract a current action based on the reaction data stored in the storage section, and compare the current action with the specific action to determine whether the specific action has been normally performed.

[0008] The action monitoring system according to the present disclosure includes the multiple occupancy sensors provided in places different from each other in the residence, a storage section, and a determination section. The determination section stores, in the storage section in advance, the specific action defined by at least the reaction order of the multiple occupancy sensors. The determination section stores reaction time as the reaction data in the storage section when any of

the multiple occupancy sensors reacts, extracts the current action based on the reaction data stored in the storage section, and compares the current action with the specific action to determine whether the specific action has been normally performed. By storing the reaction time of each occupancy sensor as the reaction data, it is possible to obtain information on from where and to where the monitoring target person has moved, and to appropriately monitor the action of the monitoring target person who moves across multiple areas.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a schematic configuration diagram of an action monitoring system according to the present embodiment.

[0010] FIG. 2 is a diagram illustrating an example of a sensor installed in each room of a residence.

[0011] FIG. 3 is a flowchart illustrating an example of data measurement processing.

[0012] FIG. 4 is a flowchart illustrating an example of specific action registration processing.

[0013] FIG. 5 is a flowchart illustrating an example of data reception processing.

[0014] FIG. 6 is a diagram illustrating an example of specific action data.

[0015] FIG. 7 is a flowchart illustrating an example of standard action registration processing.

[0016] FIG. 8 is a flowchart illustrating an example of standard action determination processing.

[0017] FIG. 9 is a flowchart illustrating an example of specific action determination processing.

DESCRIPTION OF EMBODIMENTS

[0018] Next, an embodiment of the present disclosure will be described with reference to the drawings.

[0019] FIG. 1 is a schematic configuration diagram of action monitoring system **10** of the present embodiment. As illustrated in FIG. 1, action monitoring system **10** of the present embodiment includes management server **20** that manages an entire system, and monitoring device **30** installed in each of residences A to C in which a monitoring target person lives. Residences A to C are residences in which, for example, an elderly person or a person requiring care lives alone, and include, for example, an L (living) D (dining) K (kitchen) room, a bedroom, a washroom, a bathroom, a toilet room, and an entrance as illustrated in FIG. 2. Action monitoring system **10** can be used, for example, to monitor an action of an elderly person or a person requiring care as the monitoring target person in place of a caregiver and to find out an abnormality in the action at an early stage.

[0020] Monitoring device **30** includes control section **31**, communication section **32**, operation display section **33**, speaker **34**, and sensors **40**. Control section **31** is configured as a microprocessor including CPU as a main component, and includes ROM, RAM, and the like in addition to the CPU. Operation display section **33** and speaker **34** output various information from management server **20** through display or audio. Operation display section **33** is configured as a touch panel type display section with which an operator can perform operation input.

[0021] As illustrated in FIG. 2, sensors **40** are sensors for detecting where the monitoring target person who lives in the residence is, and include occupancy sensors **41**, **42**, **43**, **44**, **45**, **46**, and **47** provided in each room, and door sensor **48** provided on an entrance door.

[0022] Occupancy sensors **41** to **47** are sensors that detect a person in a detection area in a non-contact manner, and are configured as, for example, infrared sensors that sense an infrared ray and convert the infrared ray into an electric signal. Occupancy sensors **41**, **42**, and **43** are provided in the living room, dining room, and kitchen of the LDK room, respectively. Occupancy sensor **44** is provided in the bedroom, and occupancy sensor **45** is provided in the washroom. Occupancy sensor **46** is provided in the bathroom, and occupancy sensor **47** is provided in the toilet room.

[0023] Door sensor **48** detects opening and closing of the entrance door and is configured as, for

example, a magnet-type opening/closing sensor including a permanent magnet fixed to a door side and a magnetic sensor fixed to a frame side.

[0024] Management server **20** includes processing section **21**, communication section **22**, and storage section **23**. Processing section **21** is configured as a microprocessor including CPU as a main component, and includes ROM, RAM, and the like in addition to the CPU. Communication section **22** of management server **20** is connected to communication section **32** of each monitoring device **30** via network **11** such as the Internet, and management server **20** and each monitoring device **30** exchange data and a signal with each other via communication sections **22** and **32**. Storage section **23** is configured with an HDD, an SSD, or the like, receives data measured by each monitoring device **30**, and stores the data for a certain period of time.

[0025] Next, an operation of the action monitoring system configured as described above, that is, an operation of each monitoring device **30** and an operation of management server **20** will be described. The operation of each monitoring device **30** includes data measurement processing and specific action registration processing. The operation of management server **20** includes data reception processing, standard action registration processing, standard action determination processing, and specific action determination processing.

[0026] The data measurement processing is processing of measuring (collecting) a location of the monitoring target person from a sensor provided in each room of the residence. FIG. **3** is a flowchart illustrating an example of data measurement processing executed by control section **31** of each monitoring device **30**. This processing is repeatedly executed at a predetermined time interval.

[0027] When the data measurement processing is executed, control section **31** of monitoring device **30** first determines whether occupancy sensor **41** for the living room provided in the living room reacts (step **S100**). When it is determined that occupancy sensor **41** for the living room reacts, control section **31** determines that the monitoring target person is present in the living room (step **S102**), transmits the determination result as measurement data to management server **20** (step **S136**), and ends the data measurement processing.

[0028] When it is determined in step **S100** that occupancy sensor **41** for the living room does not react, control section **31** subsequently determines whether occupancy sensor **42** for the dining room provided in the dining room reacts (step **S104**). When it is determined that occupancy sensor **41** for the dining room reacts, control section **31** determines that the monitoring target person is present in the dining room (step **S106**), transmits the determination result as measurement data to management server **20** (step **S136**), and ends the data measurement processing.

[0029] When it is determined in step **S104** that occupancy sensor **42** for the dining room does not react, control section **31** subsequently determines whether occupancy sensor **43** for the kitchen provided in the kitchen reacts (step **S108**). When it is determined that occupancy sensor **43** for the kitchen reacts, control section **31** determines that the monitoring target person is present in the kitchen (step **S110**), transmits the determination result as measurement data to management server **20** (step **S136**), and ends the data measurement processing.

[0030] When it is determined in step **S108** that occupancy sensor **43** for the kitchen does not react, control section **31** subsequently determines whether occupancy sensor **44** for the bedroom provided in the bedroom reacts (step **S112**). When it is determined that occupancy sensor **44** for the bedroom reacts, control section **31** determines that the monitoring target person is present in the bedroom (step **S114**), transmits the determination result as measurement data to management server **20** (step **S136**), and ends the data measurement processing.

[0031] When it is determined in step **S112** that occupancy sensor **44** for the bedroom does not react, control section **31** subsequently determines whether occupancy sensor **45** for the washroom provided in the washroom reacts (step **S116**). When it is determined that occupancy sensor **45** for the washroom reacts, control section **31** determines that the monitoring target person is present in the washroom (step **S118**), transmits the determination result as measurement data to management server **20** (step **S136**), and ends the data measurement processing.

[0032] When it is determined in step S116 that occupancy sensor 45 for the washroom does not react, control section 31 subsequently determines whether occupancy sensor 46 for the bathroom provided in the bathroom reacts (step S120). When it is determined that occupancy sensor 46 for the bathroom reacts, control section 31 determines that the monitoring target person is present in the bathroom (step S122), transmits the determination result as measurement data to management server 20 (step S136), and ends the data measurement processing.

[0033] When it is determined in step S120 that occupancy sensor 46 for the bathroom does not react, control section 31 subsequently determines whether occupancy sensor 47 for the toilet room provided in the toilet room reacts (step S124). When it is determined that occupancy sensor 47 for the toilet room reacts, control section 31 determines that the monitoring target person is present in the toilet room (step S126), transmits the determination result as measurement data to management server 20 (step S136), and ends the data measurement processing.

[0034] When it is determined in step S124 that occupancy sensor 47 for the toilet room does not react, control section 31 subsequently determines whether door sensor 48 for the entrance provided in the entrance door reacts (step S128). When it is determined that door sensor 48 for the entrance reacts, control section 31 determines whether it has been determined in step S134 described later that the monitoring target person has been at home (step S130). When it is determined that it has been determined that the monitoring target person has been at home, control section 31 determines that the monitoring target person has gone out (step S132), transmits the determination result as measurement data to management server 20 (step S136), and ends the data measurement processing. On the other hand, when it is determined that it has not been determined that the monitoring target person has been at home (it has been determined that the monitoring target person has been out), control section 31 determines that the monitoring target person has returned, that is, is at home (step S134), transmits the determination result as measurement data to management server 20 (step S136), and ends the data measurement processing.

[0035] The specific action registration processing is processing of registering a periodic schedule (specific action) to be executed by the monitoring target person in advance for schedule management, determination of suitability of the action of the monitoring target person, and the like. FIG. 4 is a flowchart illustrating an example of specific action registration processing executed by control section 31 of monitoring device 30. This processing is repeatedly executed at a predetermined time interval.

[0036] When the specific action registration processing is executed, control section 31 first displays a menu screen on operation display section 33 and receives selection of an action type (step S150). Then, control section 31 determines whether “dishwashing” is selected as the action type (step S152), whether “stroll” is selected as the action type (step S154), whether “garbage disposal” is selected as the action type (step S156), and whether “new” for creating a new action type is selected as the action type (step S158).

[0037] When it is determined in step S152 that “dishwashing” has been selected, control section 31 receives an input of an execution time slot in which dishwashing is performed and a time required for dishwashing (staying time in the kitchen) (step S160). Subsequently, control section 31 acquires a movement area (room) to which a person performing dishwashing moves and a movement order between the movement areas (step S162). The acquisition of the movement area and the movement order is performed by creating and registering a relationship between the action type and the movement area and movement order in advance and deriving the corresponding movement area and movement order when the action type is given. In the present embodiment, the movement area for dishwashing includes the kitchen and dining room, and the movement order for dishwashing is an order from the kitchen to the dining room, and to the kitchen. Then, control section 31 generates specific action data for dishwashing including the acquired movement area and movement order and the input execution time slot and required time (staying time) (step S178), transmits the generated specific action data to management server 20 in order to register the specific action data

in management server **20** (step **S180**), and ends the specific action registration processing.

[0038] When it is determined in step **S154** that “stroll” has been selected, control section **31** receives an input of an execution time slot in which a stroll is performed and a required time (outgoing time) (step **S164**). Subsequently, control section **31** acquires a movement area (room) to which a person strolling moves and a movement order between the movement areas (step **S166**). In the present embodiment, the movement area for a stroll includes the living room and the entrance, and the movement order for a stroll is an order from the living room to the entrance (going out), to the entrance (returning), and to the living room. Then, control section **31** generates specific action data for a stroll including the acquired movement area and movement order and the input execution time slot and required time (outgoing time) (step **S178**), transmits the generated specific action data to management server **20** in order to register the specific action data in management server **20** (step **S180**), and ends the specific action registration processing.

[0039] When it is determined in step **S156** that “garbage disposal” has been selected, control section **31** receives an input of an execution day of the week and an execution time slot in which garbage disposal is performed (step **S168**). Subsequently, control section **31** acquires a movement area (room) to which a person performing garbage disposal moves and a movement order between the movement areas (step **S170**). In the present embodiment, the movement area for garbage disposal includes the living room, the kitchen, and the entrance, and the movement order for garbage disposal is an order from the living room to the kitchen, and to the entrance. Then, control section **31** generates specific action data for garbage disposal including the acquired movement area and movement order and the input execution day of the week and execution time slot (step **S178**), transmits the generated specific action data to management server **20** in order to register the specific action data in management server **20** (step **S180**), and ends the specific action registration processing.

[0040] When it is determined in step **S158** that “new” has been selected, control section **31** receives each of an input of a movement area and a movement order (step **S172**), an input of an execution day of the week and an execution time slot (step **S174**), and an input of a required time (step **S176**). Then, control section **31** generates new specific action data including the input movement area and movement order, the execution day of the week, the execution time slot, and the required time (step **S178**), transmits the generated specific action data to management server **20** in order to register the specific action data in management server **20** (step **S180**), and ends the specific action registration processing.

[0041] Next, operations (data reception processing, standard action registration processing, standard action determination processing, and specific action determination processing) of management server **20** will be described.

[0042] The data reception processing is processing of receiving measurement data transmitted from each monitoring device **30**. FIG. 5 is a flowchart illustrating an example of data reception processing executed by processing section **21** of management server **20**. This processing is repeatedly executed at a predetermined time interval.

[0043] When the data reception processing is executed, processing section **21** of management server **20** first determines whether measurement data has been received from monitoring device **30** (step **S200**). When it is determined that measurement data has been received, processing section **21** accesses a time server via the Internet to acquire the current year, month, day, time (hour/minute/second), and day of the week as time information (step **S202**), stores the acquired time information in storage section **23** in association with the measurement data (step **S204**), and ends the data reception processing. The time information may be acquired by reading the current time from a real-time clock (RTC).

[0044] When it is determined in step **S200** that the measurement data has not been received, processing section **21** determines whether the specific action data has been received from monitoring device **30** (step **S206**). When it is determined that the specific action data has not been

received, processing section **21** ends the data reception processing. On the other hand, when it is determined that the specific action data has been received, processing section **21** registers the received specific action data in storage section **23** (step **S208**), and ends the data reception processing. An example of the specific action data is illustrated in FIG. **6**. The specific action is performed along with movement across multiple areas (rooms) of the residence, and the specific action data includes the movement order, the execution time slot, the execution day of the week, and the required time as illustrated in FIG. **6**. In the present embodiment, depending on a type of the specific action, some information (the execution day of the week and the required time) may be omitted. The specific action data is used in specific action determination processing described later. [0045] FIG. **7** is a flowchart illustrating an example of the standard action registration processing executed by processing section **21** of management server **20**. This processing is repeatedly executed at a predetermined time interval. Here, the standard action indicates a daily pattern of basic actions such as wake-up, toilet, meal, bathing, and sleep.

[0046] In the standard action registration processing, processing section **21** first determines whether the standard action is unregistered (step **S210**). When it is determined that the standard action is unregistered, processing section **21** determines whether measurement data for a certain period of time (for example, for one month or one week) has been accumulated in storage section **23** by the above-described data reception processing (step **S212**). When it is determined that the measurement data for the certain period of time has not been accumulated, processing section **21** determines that a sufficient amount of measurement data for extracting the standard action has not been accumulated, and ends the standard action registration processing. On the other hand, when it is determined that the measurement data for the certain period of time has been accumulated, processing section **21** extracts the standard action including the wake-up time, the meal time, the number of times visiting toilet, the bathing time, the bedtime, and the like from the measurement data for the certain period of time (step **S214**). For example, the extraction of the wake-up time can be performed by acquiring an exit time when an exit from the bedroom is determined after a stay in the bedroom for a predetermined time or more (a time difference between an entry time and the exit time is a predetermined time or more) is determined based on the measurement data of the bedroom and the time information associated with the measurement data. Extraction of the meal time can be performed by acquiring a stay time when a stay is determined for a predetermined time or more in the dining room after a movement from the kitchen to the dining room is determined based on the measurement data of the kitchen and the dining room and the time information associated with the measurement data. Extraction of the number of times visiting toilet can be performed by counting the number of detections each time an entry into the toilet room is detected based on measurement data of the toilet room and the time information associated with the measurement data. Extraction of the bathing time can be performed by acquiring the staying time (time difference between an entry time and an exit time) of the bathroom based on the measurement data of the bathroom and the time information associated with the measurement data. Extraction of the bedtime can be performed by acquiring an entry time into the bedroom when it is determined that the person stays in the bedroom for a predetermined time or more after the entry into the bedroom is determined based on the measurement data of the bedroom and the time information associated with the measurement data. The extraction of the wake-up time and the bedtime may be performed using a dedicated sleep sensor. When the standard action is extracted in this manner, processing section **21** registers the extracted standard action in storage section **23** (step **S216**), and ends the standard action registration processing.

[0047] When it is determined in step **S210** that the standard action has been registered, processing section **21** determines whether a season has changed based on the current year, month, and day (step **S218**). In the present embodiment, a spring season is March, April, and May, a summer season is June, July, and August, an autumn season is September, October, and November, and a winter season is December, January, and February. When it is determined that the season has

changed, processing section **21** determines whether data for a certain period of time after a season change has been accumulated in storage section **23** (step **S220**). When it is determined that the data for the certain period of time after the season change has been accumulated, processing section **21** corrects the standard action based on the data after the season change (step **S222**), updates the standard action registered in storage section **23** (step **S224**), and ends the standard action registration processing. In the present embodiment, by updating the standard action for each season, it is possible to reflect a change in the wake-up time, the bedtime, or the like, which is caused by the change in the season, in the standard action. When it is determined in step **S218** that the season has not changed or when it is determined in step **S220** that data for the certain period of time after the season change has not yet been accumulated, processing section **21** ends the standard action registration processing without updating the standard action.

[0048] FIG. **8** is a flowchart illustrating an example of a standard action determination processing executed by processing section **21** of management server **20**. This processing is repeatedly executed at a predetermined time interval.

[0049] When the standard action determination processing is executed, processing section **21** first determines whether the standard action has been registered in storage section **23** (step **S230**) and whether measurement data for one day today has been accumulated in storage section **23** (step **S232**). When it is determined that the standard action has not been registered or it is determined that the measurement data for one day today has not been accumulated, processing section **21** ends the standard action determination processing. On the other hand, when it is determined that the standard action has been registered and the measurement data for one day today has been accumulated, processing section **21** extracts a current action including the wake-up time, the meal time, the number of times visiting toilet, the bathing time, the bedtime, and the like from the measurement data for one day today (step **S234**). The current action is extracted from the data for one day today, but may be extracted from data for several days in the past from today.

[0050] Next, processing section **21** compares the extracted current action with the standard action registered in storage section **23** (step **S236**), and determines whether the current action matches the standard action within a predetermined range (step **S238**). This determination is performed by determining whether a difference between the current action and the standard action (a difference in time or a difference in the number of times) is within an allowable range. When it is determined that the current action matches the standard action within the predetermined range, processing section **21** determines that the current action is normal (step **S240**), and ends the standard action determination processing. On the other hand, when it is determined that the current action does not match the standard action within the predetermined range, processing section **21** determines that the current action is abnormal (step **S242**), and ends the standard action determination processing. Accordingly, it is possible to detect a sign of an abnormal change different from the standard action at an early stage, which contributes to early treatment. In addition, since the standard action is appropriately updated based on the action that changes due to a season factor, it is possible to prevent the current action from being erroneously determined to be abnormal.

[0051] FIG. **9** is a flowchart illustrating an example of specific action determination processing executed by processing section **21** of management server **20**. This processing is repeatedly executed at a predetermined time interval.

[0052] When the specific action determination processing is executed, processing section **21** first determines whether the specific action has been registered in storage section **23** (step **S250**). When it is determined that the specific action has not been registered, processing section **21** ends the specific action determination processing. On the other hand, when it is determined that the specific action has been registered, processing section **21** determines whether an execution time slot in which a specific action as a determination target among the registered specific actions is to be executed has elapsed (step **S252**). This determination is performed by determining whether the execution time slot of the execution day of the week has elapsed when the execution day of the

week has been determined for the specific action as the determination target. When it is determined that the execution time slot has not elapsed, processing section **21** ends the specific action determination processing.

[0053] On the other hand, when it is determined that the execution time slot has elapsed, processing section **21** extracts the movement order of the room (action order) in the execution time slot from the measurement data within the execution time slot stored in storage section **23**, and checks whether the extracted movement order matches the movement order of the room (action order) in the specific action as the determination target (step **S254**). This processing is performed by extracting measurement data (rooms in which the monitoring target person is present) within the execution time slot in time series, and determining whether a data group matching the action order of the specific action as the determination target is included in the extracted measurement data.

[0054] Subsequently, processing section **21** determines whether an execution day of the week has been registered in the specific action as the determination target (step **S256**). When it is determined that the execution day of the week has been registered, processing section **21** checks whether the day of the week today matches the execution day of the week of the specific action as the determination target (step **S258**). On the other hand, when it is determined that the execution day of the week has not been registered, processing section **21** skips step **S258**.

[0055] Next, processing section **21** determines whether the required time has been registered in the specific action as the determination target (step **S260**). When it is determined that the required time has been registered, processing section **21** calculates the staying time based on the time information associated with the corresponding measurement data within the execution time slot, and determines whether the calculated execution time and the required time of the specific action as the determination target match within a predetermined range (step **S262**). This determination is performed, for example, by comparing the staying time in the kitchen with the time required for dishwashing when the specific action as the determination target is dishwashing. In addition, the determination is performed by comparing the outgoing time and the required time of the stroll when the specific action as the determination target is the stroll. On the other hand, when it is determined that the required time has not been registered, processing section **21** skips step **S262**.

[0056] When it is determined that all of the movement order, the execution day of the week, and the required time match the specific action as the determination target (“YES” in step **S264**), processing section **21** determines that the specific action as the determination target is executed (step **S266**), transmits the determination result to monitoring device **30** (step **S268**), and ends the specific action determination processing. Upon receiving the determination result indicating that the specific action has been executed, monitoring device **30** displays information indicating the execution state of the specific action or the like on operation display section **33**. Accordingly, it is possible to prompt the monitoring target person to continuously execute the specific action.

[0057] On the other hand, when it is determined that any one of the movement order, the execution day of the week, and the required time does not match the specific action as the determination target (“NO” in step **S264**), processing section **21** determines that the specific action as the determination target has not been executed (step **S270**), transmits the determination result to monitoring device **30** (step **S272**), and ends the specific action determination processing. Upon receiving the determination result indicating that the specific action has not been executed, monitoring device **30** displays information indicating that the specific action has not been executed or a message prompting the monitoring target person to execute the specific action on operation display section **33**. Accordingly, it is possible to cause the monitoring target person to more reliably execute the specific action.

[0058] Note that processing section **21** may totalize the presence or absence of the execution of the specific action for each predetermined period (for example, one month or one week) to calculate an execution rate, and determine that the specific action has not been appropriately executed when the execution rate is smaller than a predetermined value. Thus, the execution state of the specific action

can be obtained. In addition, processing section **21** may transmit the determination result to a portable information terminal such as a smartphone registered in advance in order to notify the monitoring target person or a protector of the determination result for the specific action. In addition, the execution time slot, the execution day of the week, and the required time which are different for each timing (for example, for each week, for each month, and for each season) may be registered as the specific action, and it may be determined whether the specific action is executed in the execution day of the week, the execution time slot, and the required time corresponding to the current timing.

[0059] Here, a correspondence relationship between main elements of the embodiment and main elements of the present disclosure described in Claims will be described. That is, occupancy sensors **41** to **47** of the present embodiment correspond to occupancy sensors of the present disclosure, storage section **23** corresponds to a storage section, and processing section **21** corresponds to a determination section. Operation display section **33** corresponds to a notification section.

[0060] It is needless to say that the present disclosure is not limited in any way to the embodiment described above, and the present disclosure can be embodied in various aspects as long as the aspects fall within the technical scope of the present disclosure.

[0061] For example, in the above-described embodiment, the specific action is registered in storage section **23** by inputting the action type (movement area and movement order), the execution time slot, the required time (staying time), and the like in advance with operation display section **33**. However, similarly to the standard action, processing section **21** may extract the specific action based on the measurement data for the certain period of time that has been acquired from monitoring device **30** and accumulated.

[0062] An action monitoring system according to the present disclosure described above includes multiple occupancy sensors provided in places different from each other in the residence, a storage section, and a determination section. The determination section stores, in the storage section in advance, a specific action defined by at least a reaction order of the multiple occupancy sensors. The determination section stores reaction time as reaction data in the storage section when any of the multiple occupancy sensors reacts, extracts a current action based on the reaction data stored in the storage section, and compares the current action with the specific action to determine whether the specific action has been normally performed. By storing the reaction time of each occupancy sensor as reaction data, it is possible to obtain information on from where and to where the monitoring target person has moved, and to appropriately monitor the action of the monitoring target person who moves across multiple areas.

[0063] The action monitoring system of the present disclosure may further include a notification section configured to notify information reflecting a determination result of the determination section. In this way, it is possible to prompt the monitoring target person to perform the specific action.

[0064] In the action monitoring system of the present disclosure, the specific action may be defined further by any of a timing, a day of the week, a time slot, and a staying time. In this way, it is possible to more accurately define the specific action, and it is possible to accurately determine whether the specific action has been performed. The timing may include a year, a month, a day, and a season.

[0065] Furthermore, in the action monitoring system of the present disclosure, the determination section may further extract a standard action based on the reaction data accumulated for a certain period of time in storage section, compare the current action with the standard action to determine whether the standard action has been normally performed, and change the standard action based on a timing change in the reaction data. In this way, even when the action of the monitoring target person changes because of a season factor or the like, it is possible to accurately determine whether the standard action has been performed.

[0066] The present disclosure is not limited to a form of an action monitoring system, and may be a form of an action monitoring method.

INDUSTRIAL APPLICABILITY

[0067] The present disclosure can be used in a manufacturing industry of an action monitoring system.

REFERENCE SIGNS LIST

[0068] **10** action monitoring system, **11** network, **20** management server, **21** processing section, **22** communication section, **23** storage section, **30** monitoring device, **31** control section, **32** communication section, **33** operation display section, **34** speaker, **40** sensors, **41** to **47** occupancy sensor, **48** door sensor.

Claims

1. An action monitoring system configured to monitor an action of a monitoring target person living in a residence, the action monitoring system comprising: multiple occupancy sensors provided in places different from each other in the residence, the multiple occupancy sensors being configured to detect the monitoring target person; a storage section configured to store data; and a determination section configured to store, in the storage section in advance, a specific action defined by at least a reaction order of the multiple occupancy sensors, store a reaction time as reaction data in the storage section when any of the multiple occupancy sensors reacts, extract a current action based on the reaction data stored in the storage section, and compare the current action with the specific action to determine whether the specific action has been normally performed.
 2. The action monitoring system according to claim 1, further comprising: a notification section configured to notify information reflecting a determination result of the determination section.
 3. The action monitoring system according to claim 1, wherein the specific action is defined further by any of a timing, a day of the week, a time slot, and a staying time.
 4. The action monitoring system according to claim 1, wherein the determination section is configured to further extract a standard action based on the reaction data accumulated for a certain period of time in the storage section, compare the current action with the standard action to determine whether the standard action has been normally performed, and change the standard action based on a timing change in the reaction data.
 5. An action monitoring method of monitoring an action of a monitoring target person living in a residence, the action monitoring method comprising: a detection step of detecting the monitoring target person in places different from each other in the residence; a first storage step of storing a specific action defined by at least a reaction order in multiple places in the detection step, in advance; a second storage step of storing a reaction time as reaction data when there is a reaction in any of the multiple places in the detection step; and a determination step of extracting a current action based on the reaction data stored in the second storage step and comparing the current action with the specific action to determine whether the specific action has been normally performed.
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